

1. Objective

The AIME 2024 subset consists of 15 challenging competition problems. The goal was to evaluate two small-scale mathematical reasoning models (Qwen2.5-Math-1.5B and DeepSeek-R1-Distill-Qwen-1.5B) using three prompting strategies:

- **Chain-of-Thought (CoT)** -- step-by-step reasoning
- **Least-to-Most (LtM)** -- decomposition into subproblems
- **Self-Consistency (SC, k=5)** -- majority voting over five CoT generations

The outputs serve as the foundation for accuracy computation, response length analysis, and error classification performed by member 6.

2. Input Data

The dataset was provided by member 1 in a unified JSONL format:

- **File:** unified_aime2024_i_official_20260311_191357.jsonl
- **Structure:** each entry contains problem_id, question, answer_normalized (the official three-digit integer answer), and metadata.

All 15 AIME 2024 problems were included, with problem texts manually imported from a local CSV (as described in the preprocessing pipeline).

3. Experimental Setup

3.1 Models

Model Name	Hugging Face Identifier
Qwen2.5-Math-1.5B	Qwen/Qwen2.5-Math-1.5B
DeepSeek-R1-Distill-Qwen-1.5B	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B

Both models were loaded in half-precision (torch.float16) and automatically placed on the available GPU using device_map="auto". Generation parameters were:

- max_new_tokens = 512
- temperature = 0.7
- top_p = 0.9
- do_sample = True (necessary for self-consistency)

3.2 Prompting Module

The prompt.py module (developed by member 2) was used. It provides the unified function run_prompt_method(model, tokenizer, question, method), which returns:

- method: the prompting strategy used
- raw_output: the generated text (or list of texts for SC)
- final_answer: the extracted numeric answer (or None if extraction failed)

Answer extraction relies on a regex looking for Answer: <number> at the end of the output.

3.3 Execution Environment

Experiments were run on **Google Colab** with a Tesla T4 GPU (16 GB VRAM) to ensure sufficient speed and stability. The entire process took approximately 1.5 hours.

4. Execution Process

A custom script `run_aime_experiments.py` was written to orchestrate the experiment. The script:

1. Loaded the dataset from the provided JSONL file.
2. For each model:
 - Loaded the tokenizer and model.
 - Iterated over all 15 problems.
 - For each problem, called `run_prompt_method` for CoT, LtM, and SC.
 - Recorded the result, including the generated output, extracted answer, ground truth, generation time, and status (success / no answer / error).
 - Wrote each record immediately to a JSONL file (one file per model) to avoid data loss.
3. Saved both result files inside a `aime_results/` folder on Google Drive.

The script is designed to be robust: any exception during generation is caught, logged with an error message, and the loop continues. For self-consistency, the five sampled outputs are concatenated with separators for readability.

5. Output Structure

Each result file (e.g., `Qwen2.5-Math-1.5B_aime_results.jsonl`) contains one JSON object per (model, problem, method) combination. Each object includes:

Field	Description
<code>model</code>	Model short name (e.g., Qwen2.5-Math-1.5B)
<code>problem_id</code>	Unique identifier from the dataset
<code>question</code>	Full problem text
<code>ground_truth</code>	Official three-digit answer
<code>method</code>	Prompting strategy used
<code>final_answer</code>	Extracted answer (or None)
<code>raw_output</code>	The model's full output (for SC, a concatenated string of 5 outputs)
<code>generation_time_sec</code>	Time taken for the generation (for SC, total time for 5 runs)

Field	Description
status	success, no_answer_extracted, or error
error_msg	Present only if status is error

These files are directly consumable by member 6 for subsequent evaluation.

6. Deliverables

The following files were uploaded to the team's GitHub repository under experiments/results/:

- Qwen2.5-Math-1.5B_aime_results.jsonl
- DeepSeek-R1-Distill-Qwen-1.5B_aime_results.jsonl

Member 6 can download them to perform accuracy calculation, response length analysis, and error classification.

7. Observations and Notes

- Model behavior: Both models occasionally produced outputs that did not follow the exact Answer: <number> format, leading to extraction failures (status no_answer_extracted). These cases are flagged for manual review. Member 6 will apply a more flexible answer extraction method to recover the intended numeric answers.
- Self-consistency: The majority vote sometimes resulted in no answer when fewer than three answers matched; this is reflected in the final_answer field.
- Long reasoning: AIME problems occasionally triggered long chains of reasoning; the 512 token limit was sufficient for all runs.
- Execution reliability: No runtime errors occurred during the execution; all 15 problems $\times$ 3 methods $\times$ 2 models completed successfully.

8. Future Integration

The results will be combined with the MATH-500 and GSM8K experiments (performed by members 3 and 4) in the final report. Member 6 will contribute error analysis and statistical plots, and member 7 will integrate everything into the final report and presentation.